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CPU Scheduling Analyzer

Simulate, Analyze, and Compare Algorithms

Number of Processes: 4 Generate Table

PROCESS	ARRIVAL TIME	BURST TIME
P1	0	5
P2	1	3
P3	2	5
P4	3	3

(•) Single Algorithm ( ) Compare All

Algorithm: Round Robin

Quantum: 3

▶ Run Simulation

Algorithm Explanation

Round Robin: Every process gets a small, equal amount of CPU time (Quantum). If a process doesn't finish within its quantum, it is paused and sent to the back of the queue. Great for responsiveness.

Execution Results

Avg WT: 6.50 Avg TAT: 10.50

PROCESS	AT	BT	CT	TAT	WT
P1	0	5	14	14	9
P2	1	3	6	5	2
P3	2	5	16	14	9
P4	3	3	12	9	6

Gantt Chart

P1	P2	P3	P4	P1	P3

Execution Log

```
[Time 0] P1 executes for 3 units.
[Time 3] P2 executes for 3 units.
[Time 6] P3 executes for 3 units.
[Time 9] P4 executes for 3 units.
[Time 12] P1 executes for 2 units.
[Time 14] P3 executes for 2 units.
```

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CPU Scheduling Analyzer

Simulate, Analyze, and Compare Algorithms

Number of Processes:

4 Generate Table

PROCESS	ARRIVAL TIME	BURST TIME	PRIORITY
P1	0	5	1
P2	1	3	2
P3	2	5	3
P4	3	3	4

◉ Single Algorithm ◯ Compare All

Algorithm:

Priority (Preemptive)

▶ Run Simulation

Algorithm Explanation

Priority (Preemptive): CPU executes the highest priority process available. If a new process arrives with a higher priority than the running one, the current process is interrupted.

Execution Results

Avg WT: 5.00 Avg TAT: 9.00

PROCESS	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	3	8	7	4
P3	2	5	13	11	6
P4	3	3	16	13	10

Gantt Chart

Execution Log

```
[Time 0] P1 executes.
[Time 5] P2 executes.
[Time 8] P3 executes.
[Time 13] P4 executes.
```

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CPU Scheduling Analyzer

Simulate, Analyze, and Compare Algorithms

Number of Processes:

4 Generate Table

PROCESS	ARRIVAL TIME	BURST TIME	PRIORITY
P1	0	5	1
P2	1	3	2
P3	2	5	3
P4	3	3	4

◉ Single Algorithm ◯ Compare All

Algorithm:

Priority (Non-Preemptive)

▶ Run Simulation

Algorithm Explanation

Priority (Non-Preemptive): CPU is allocated to the process with the highest priority (lowest number). Once started, the process runs to completion regardless of new arrivals.

Execution Results

Avg WT: 5.00 Avg TAT: 9.00

PROCESS	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	3	8	7	4
P3	2	5	13	11	6
P4	3	3	16	13	10

Gantt Chart

Execution Log

```
[Time 0] P1 selected (Priority: 1).
[Time 5] P2 selected (Priority: 2).
[Time 8] P3 selected (Priority: 3).
[Time 13] P4 selected (Priority: 4).
```

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Simulate, Analyze, and Compare Algorithms

Number of Processes:

4 Generate Table

PROCESS	ARRIVAL TIME	BURST TIME
P1	0	5
P2	1	3
P3	2	5
P4	3	3

Single Algorithm Compare All

Algorithm:

SRTF (Preemptive SJF)

▶ Run Simulation

Algorithm Explanation

SRTF (Shortest Remaining Time First): The preemptive version of SJF. If a new process arrives with a shorter burst time than what is currently left on the running process, the CPU switches to the new, shorter job.

Execution Results

Avg WT: 4.00 Avg TAT: 8.00

PROCESS	AT	BT	CT	TAT	WT
P1	0	5	11	11	6
P2	1	3	4	3	0
P3	2	5	16	14	9
P4	3	3	7	4	1

Gantt Chart

Execution Log

```
[Time 0] P1 executes.
[Time 1] P2 executes.
[Time 4] P4 executes.
[Time 7] P1 executes.
[Time 11] P3 executes.
```

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CPU Scheduling Analyzer

Simulate, Analyze, and Compare Algorithms

Number of Processes:

4 Generate Table

PROCESS	ARRIVAL TIME	BURST TIME
P1	0	5
P2	1	4
P3	2	5
P4	3	3

Single Algorithm Compare All

Algorithm:

SJF (Non-Preemptive)

▶ Run Simulation

Algorithm Explanation

SJF (Shortest Job First): Looks at all currently available processes and executes the one with the shortest burst time. This mathematically minimizes the average waiting time for the current queue, but it is non-preemptive.

Execution Results

Avg WT: 4.75 Avg TAT: 9.00

PROCESS	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	4	12	11	7
P3	2	5	17	15	10
P4	3	3	8	5	2

Gantt Chart

Execution Log

```
[Time 0] P1 selected (Shortest Burst).
[Time 5] P4 selected (Shortest Burst).
[Time 8] P2 selected (Shortest Burst).
[Time 12] P3 selected (Shortest Burst).
```

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CPU Scheduling Analyzer

Simulate, Analyze, and Compare Algorithms

Number of Processes:

4 Generate Table

PROCESS	ARRIVAL TIME	BURST TIME
P1	0	5
P2	1	4
P3	2	5
P4	3	3

(•) Single Algorithm ( ) Compare All

Algorithm:

FCFS

▶ Run Simulation

Algorithm Explanation

FCFS (First Come First Serve): The simplest scheduling algorithm. Processes are executed exactly in the order they arrive. It is non-preemptive, meaning once a process starts, it cannot be interrupted.

Execution Results

Avg WT: 5.50 Avg TAT: 9.75

PROCESS	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	1	4	9	8	4
P3	2	5	14	12	7
P4	3	3	17	14	11

Gantt Chart

Execution Log

```
[Time 0] P1 starts.
[Time 5] P1 finished.
[Time 5] P2 starts.
[Time 9] P2 finished.
[Time 9] P3 starts.
[Time 14] P3 finished.
[Time 14] P4 starts.
[Time 17] P4 finished.
```